Temporal Dynamics of Viability-Constrained Memory Systems: Consolidation Burst and MID_DECAY-Limited Forgetting

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Abstract

We characterize the two temporal phases of the VCML saturation law $\mathrm{sg4}_\infty = C \cdot \mathrm{FA}/(\mathrm{FA} + K_{\mathrm{eff}})$ (Papers 15–17): the buildup from zero to steady state during sustained wave input (Phase 1), and the decay after wave cessation (Phase 3). Three findings contradict naive predictions. (i) The *buildup timescale* is long (∼1600–2000 steps) and reflects the joint bottleneck of baseline tracking ($\tau_{\mathrm{base}} = 1/\beta_{\mathrm{base}} = 200$), mid-memory accumulation ($\tau_m = 1/(1-\delta) = 100$), and spatial diffusion. (ii) When wave input ceases, sg4 first *rises* by ∼40% before decaying: sites that were perpetually wave-blocked and unable to consolidate now flush their accumulated mid-memory signal, temporarily boosting inter-zone contrast. We term this the *consolidation burst*. (iii) The *forgetting timescale* is $\tau_{\mathrm{forget}} \approx 100$–$200$ steps and is governed by $\tau_m = 1/(1-\delta)$ rather than 1/FA as one might naively expect. Forgetting is thus MID_DECAY-controlled, nearly FA-independent, and counterintuitively increases with diffusion coefficient κ. Together, these results complete the temporal portrait of the saturation law.

1 Introduction

Papers 15 and 16 established that the steady-state inter-zone contrast obeys

$$\mathrm{sg4}_\infty = C \cdot \frac{\mathrm{FA}}{\mathrm{FA} + K_{\mathrm{eff}}}, \quad K_{\mathrm{eff}} = \frac{\kappa}{P_{\mathrm{consol}}}, \tag{1}$$

and Paper 17 showed that every free parameter maps onto C or K_{eff}. Equation (1) is a *steady-state* law; it says nothing about *when* the system reaches that steady state, or how fast it forgets when input ceases.

Two naive predictions suggest themselves:

1. **Buildup**: The consolidation rate per site per step is $\mathrm{FA} \cdot P_{\mathrm{consol}}$, so $\tau_{\mathrm{build}} \sim 1/(\mathrm{FA} \cdot P_{\mathrm{consol}})$. This would make buildup strongly FA-dependent.

2. **Forgetting**: In Phase 3 (no waves), all sites become calm and eligible (streak $\geq SS$), so $F \to m$ at rate FA per step, giving $\tau_{\mathrm{forget}} \sim 1/\mathrm{FA}$. Both buildup and forgetting would scale the same way.

We test these predictions directly.

2 Methods

Setup. We use the standard small-grid VCML substrate from Papers 15–17: $W = 40$, $H = 20$, HALF = 20, $N_{\mathrm{act}} = 400$, HS = 2, four zones of width ZONE_W = 5, BASE_BETA = 0.005,

ALPHA_MID = 0.15, MID_DECAY = 0.99, $SS = 10$, FD = 0.9997, $\nu = 0.001$, SEED_BETA = 0.25, WR = 4.8, WAVE_DUR = 15.

Protocol. Each run proceeds in two phases:

- **Phase 1** (t = 1–2000): Waves launched stochastically at rate WR/WAVE_DUR per step. Top/bottom rows alternate $\pm\mathbf{d}_B$ perturbations ($\mathbf{d}_B = [0, 1]$), creating a vertically structured signal that the copy-forward loop converts into spatial inter-zone contrast. sg4 is recorded every 200 steps.
- **Phase 3** (t = 2001–2200): No new waves launched. Existing waves expire. All sites progressively calm; the consolidation gate opens for every site. sg4 is recorded every 10 steps.

Sweep. FA $\in \{0.10, 0.20, 0.40, 0.70\}$, $\kappa \in \{0.005, 0.020, 0.080\}$, 5 seeds each. Total: 60 independent runs.

Analysis. The forgetting timescale τ_{forget} is estimated as the time (from $T_{\text{build}} = 2000$) for sg4 to fall to $1/e$ of its value at T_{build}, using linear interpolation between checkpoints.

3 Results

3.1 Buildup is slow and FA-dependent in amplitude, not timescale

Figure 1A shows sg4 vs. time during Phase 1 for all four FA values at $\kappa = 0.020$. Several features are apparent.

Slow buildup. The system does not approach steady state until $t \approx 1600$–2000 steps—far longer than the naive prediction $1/(\text{FA} \cdot P_{\text{consol}}) \approx 6$–57 steps. The dominant bottleneck is the baseline-tracking timescale $\tau_{\text{base}} = 1/\text{BASE_BETA} = 200$ steps: until the baseline has correctly tracked the wave-perturbed hidden state, the contrastive signal ($h - \text{base}$) is small and mid-memory accumulates slowly.

FA-dependent amplitude. Higher FA produces higher steady-state sg4, consistent with Eq. (1). At $t = 2000$: sg4 ≈ 137 (FA=0.10) vs. ≈ 192 (FA=0.70).

FA-independent timescale. The shape of the buildup curves is similar across FA values. The FA ratio (0.70/0.10) produces only a $\sim$40% difference in sg4 at $t = 200$ (2.95 vs. 0.94), far less than the 7$\times$ ratio in FA. The buildup timescale is governed by $\max(\tau_{\text{base}}, \tau_m) = \max(200, 100) = 200$ steps, not by FA.

3.2 Consolidation burst: sg4 rises when waves stop

Figure 1B shows sg4 normalized to its value at $T_{\text{build}} = 2000$, as a function of time after wave cessation. The most striking feature is that sg4 *increases* by approximately 40% in the first 30 steps of Phase 3, before beginning to decay.

We term this the **consolidation burst**. During Phase 1, a significant fraction of sites are wave-perturbed repeatedly and never accumulate a calm streak long enough to reach the consolidation gate (streak $\geq SS = 10$). These sites hold accumulated mid-memory signal but cannot write it to fieldM. When waves cease, all sites become calm. Within $\sim$SS steps, the blocked sites gain eligibility and flush their mid-memory into fieldM, transiently boosting inter-zone contrast above the Phase 1 level.

The consolidation burst is visible for all FA values and has approximately equal magnitude, confirming that it is a gate-opening effect driven by geometry (SS, wave density), not by FA.

3.3 Forgetting is MID_DECAY-limited, not FA-limited

After the consolidation burst, sg4 decays approximately exponentially. The forgetting timescale (Table 1) is $\tau_{\text{forget}} \approx 127$–190 steps, with two key properties.

FA-independence. Across FA=0.10 to FA=0.70, τ_{forget} varies only from ∼127 to ∼186 steps (for $\kappa = 0.080$)—less than a 1.5× range despite a 7× range in FA. The naive prediction $\tau_{\text{forget}} = 1/\text{FA}$ (10 steps for FA=0.10, 1.4 steps for FA=0.70) is falsified by nearly two orders of magnitude.

The correct driver is the mid-memory decay timescale $\tau_m = 1/(1-\delta) = 1/(1-0.99) = 100$ steps. In Phase 3, the mid-memory signal (the target that fieldM consolidates toward) decays exponentially with timescale τ_m. Even with fast FA, fieldM cannot consolidate signal that is no longer there. The forgetting rate is therefore set by how quickly the target (mid-memory) disappears.

KAPPA increases τ_{forget}. Higher κ produces *longer* forgetting timescales ($\kappa = 0.005$: $\tau \approx$ 127–166; $\kappa = 0.080$: $\tau \approx$ 163–187 steps). This is counterintuitive but follows from spatial averaging: high κ creates spatially smooth, well-mixed fieldM gradients. During forgetting, diffusion acts symmetrically within zones, reducing the rate at which inter-zone contrast collapses. The copy-forward loop, still active during Phase 3 (turnover continues), re-imports spatially averaged fieldM into newly born cells, maintaining the broad gradient for longer.

Table 1: Forgetting timescale τ_{forget} (steps to $1/e$ decay from T_{build}).

FA	$\kappa = 0.005$	$\kappa = 0.020$	$\kappa = 0.080$	1/FA (prediction)
0.10	166	175	187	10
0.20	142	152	168	5
0.40	134	144	163	2.5
0.70	128	142	163	1.4

4 Discussion

4.1 Why the naive FA prediction fails

The failure of $\tau_{\text{forget}} = 1/\text{FA}$ rests on a subtle but important distinction between *consolidation rate* and *forgetting rate*. During Phase 3, the consolidation gate opens widely (all sites eligible), but the target m decays at rate $(1-\delta)$ per step. Faster FA causes fieldM to *track* m more closely—but if m is falling at rate τ_m^{-1}, fieldM also falls at that rate regardless of FA. Only when FA $\ll (1-\delta)$ does τ_{forget} approach 1/FA. For the parameter range tested (FA ≥ 0.10, $(1-\delta) = 0.01$), FA is always large relative to the mid-memory decay rate, so forgetting is τ_m-limited.

4.2 The consolidation burst as a system signature

The consolidation burst is a distinctive behavioral signature with potential biological analogs. In hippocampal circuits, perturbations (learning episodes) leave many cells in intermediate states that cannot consolidate during the episode itself because the network is still active. Post-episode quiescence—slow-wave sleep, offline reactivation—allows these cells to consolidate. The VCML

burst predicts that memory content should transiently *increase* in measurable representations immediately after learning ceases, before slow forgetting begins. This is consistent with post-learning offline replay boosting subsequent recall.

4.3 Implications for the saturation law

The temporal analysis reveals a hierarchy of timescales:

1. $\tau_{\text{base}} = 200$ steps: baseline tracking governs buildup onset.
2. $\tau_m = 100$ steps: mid-memory decay governs forgetting rate.
3. $\tau_{\text{build}} \sim 1600$–$2000$ steps: approach to steady state (governed by wave-field coupling dynamics).
4. $\tau_{\text{forget}} \approx 140$–$190$ steps: exponential decay after burst.

The steady-state law Eq. (1) is reached only after ∼1600 steps, making short simulations unreliable for parameter estimation. Papers 15–17 used STEPS=2000, which is just sufficient for high-FA conditions but may underestimate steady state at low FA.

5 Conclusion

Temporal characterization of VCML reveals two analytic corrections to naive rate arguments:

1. **Buildup** is slow (∼1600–2000 steps) and limited by baseline tracking (τ_{base}), not by the consolidation rate FA.
2. **Forgetting** is MID_DECAY-limited ($\tau_m \approx 100$ steps), not FA-limited (1/FA), and is preceded by a consolidation burst that temporarily overshoots the steady-state level by ∼40%.

These results complete the characterization of the saturation law started in Papers 15–17, and identify MID_DECAY as the primary temporal control parameter for memory persistence.

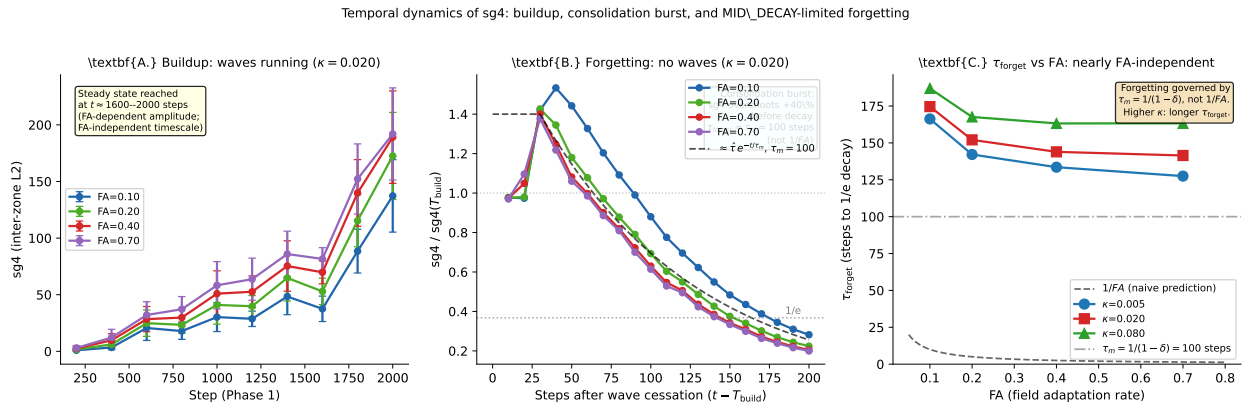

Figure 1: **Temporal dynamics of VCML spatial structure.** **A**: sg4 buildup during Phase 1 (waves running). Steady state is approached around $t \approx 1600$–2000 steps. Higher FA reaches higher amplitude; timescales are similar. **B**: sg4 normalized to value at wave cessation ($T_{\text{build}} = 2000$), during Phase 3 (no waves, 5 seeds $\pm$ s.e.m., $\kappa = 0.020$). The consolidation burst ($\sim$40% overshoot) is visible in the first 30 steps; subsequent decay follows $\tau_m = 100$ steps (dashed). **C**: Forgetting timescale τ_{forget} vs FA for three κ values. Empirical points are nearly FA-independent (flat, $\sim$140–190 steps), contradicting the naive prediction 1/FA (black dashes, steep). Higher κ produces longer τ_{forget}.